

MATH 161, Autumn 2025
SCRIPT 1: Sets, Functions and Cardinality

Sets and functions are among the most fundamental objects in mathematics. A formal treatment of set theory was first undertaken at the end of the 19th Century and was finally codified in the form of the Zermelo-Fraenkel axioms. While fascinating in its own right, pursuit of these formalisms at this point would distract us from our main purpose of studying Calculus. Thus, we present a simplified version that will suffice for our immediate purposes.

Sets

Definition 1.1. (Working Definition) A *set* is an object S with the property that, given any x , we have the dichotomy that precisely one of the two conditions $x \in S$ or $x \notin S$ is true. In the former case, we say that x is an *element* of S , and in the latter, we say that x is not an element of S .

A set is often presented in one of the following forms:

- A complete listing of its elements.

Example: the set $S = \{1, 2, 3, 4, 5\}$ contains precisely the five smallest positive integers.

- A listing of some of its elements with ellipses to indicate unnamed elements.

Example 1: the set $S = \{3, 4, 5, \dots, 100\}$ contains the positive integers from 3 to 100, including 6 through 99, even though these latter are not explicitly named.

Example 2: the set $S = \{2, 4, 6, \dots, 2n, \dots\}$ is the set of all positive even integers.

- A two-part indication of the elements of the set by first identifying the source of all elements and then giving additional conditions for membership in the set.

Example 1: $S = \{x \in \mathbb{N} \mid x \text{ is prime}\}$ is the set of primes.

Example 2: $S = \{x \in \mathbb{Z} \mid x^2 < 3\}$ is the set of integers whose squares are less than 3.

Definition 1.2. Two sets A and B are *equal* if they contain precisely the same elements, that is, $x \in A$ if and only if $x \in B$. When A and B are equal, we denote this by $A = B$.

Definition 1.3. A set A is a *subset* of a set B if every element of A is also an element of B , that is, if $x \in A$, then $x \in B$. When A is a subset of B , we denote this by $A \subset B$. If $A \subset B$ but $A \neq B$ we say that A is a *proper* subset of B .

Exercise 1.4. Let $A = \{1, \{2\}\}$. Is $1 \in A$? Is $2 \in A$? Is $\{1\} \subset A$? Is $\{2\} \subset A$? Is $1 \subset A$? Is $\{1\} \in A$? Is $\{2\} \in A$? Is $\{\{2\}\} \subset A$? Explain.

Set A contains elements 1 and $\{2\}$, with the number 1 and the set of 2 being inside of Set A .

Proof of (a). Element 1 is within subset A . Therefore, we can say Set A contains the element 1. Hence, we can state $1 \in A$.

□

Proof of (b). Element 2 is not within subset A. This is because Set A contains {2}, and not the element 2 itself. Therefore, $2 \notin A$. □

Proof of (c). {1} is a subset of A. As defined in **Definition 1.3**, we can say that {1} is a subset because every element is present in Set A. □

Proof of (d). {2} is not a subset of A. This is because the element 2 is not present in A. Instead, only {2} is present. Therefore, $\{2\} \not\subset A$. □

Proof of (e). 1 is not a subset of A. This is because 1 is not a set; therefore, it may not be a subset. □

Proof of (f). {1} is not in A. This is because only the element 1 is in Set A and not the set itself. Therefore, $\{1\} \notin A$. □

Proof of (g). {2} is in A. This is because Set A contains the set of 2 itself. Therefore, $\{2\} \in A$. □

Proof of (h). {{2}} is a subset of A. This is because Set {{2}} contains the element {2}, which is present in A. Therefore, $\{\{2\}\} \subset A$. □

Definition 1.5. Let A and B be two sets. The *union* of A and B is the set

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

Definition 1.6. Let A and B be two sets. The *intersection* of A and B is the set

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Theorem 1.7. *no proof required* Let A and B be two sets. Then:

a) $A = B$ if and only if $A \subset B$ and $B \subset A$.

b) $A \subset A \cup B$.

c) $A \cap B \subset A$.

A special example of the intersection of two sets is when the two sets have no elements in common. This motivates the following definition.

Definition 1.8. The *empty set* is the set with no elements, and it is denoted $\emptyset$. That is, no matter what x is, we have $x \notin \emptyset$.

Definition 1.9. Two sets A and B are *disjoint* if $A \cap B = \emptyset$.

Exercise 1.10. Show that if A is any set, then $\emptyset \subset A$.

Proof. Let $\emptyset$ be a set that is defined by the domain

$$\emptyset = \{\} \quad (1.0.1)$$

where the set contains no elements.

By *Definition 1.3*, $A \subset B$ if for all $x \in A$ there is an $x \in B$.

As the $\emptyset$ contains no elements, we can therefore state that every element in the $\emptyset$ is present in Set A. Therefore, we may state that $\emptyset \subset A$ regardless if A is any set. □

Definition 1.11. Let A and B be two sets. The *difference* of B from A is the set

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

The set $A \setminus B$ is also called the *complement* of B relative to A . When the set A is clear from the context, this set is sometimes denoted B^c , but we will try to avoid this imprecise formulation and use it only with warning.

Exercise 1.12. Let $A = \{x \in \mathbb{N} \mid x \text{ is even}\}$; $B = \{x \in \mathbb{N} \mid x \text{ is odd}\}$; $C = \{x \in \mathbb{N} \mid x \text{ is prime}\}$; $D = \{x \in \mathbb{N} \mid x \text{ is a perfect square}\}$. Find all possible set differences.

Answer. The following are all the possible set differences

- a.) $A \setminus B = A$
- b.) $B \setminus A = B$
- c.) $A \setminus C = \{x \in \mathbb{N} \mid x \text{ is even and } x \geq 4\}$
- d.) $A \setminus C = \{x \in \mathbb{N} \mid x \text{ is even and } x \geq 4\}$
- e.) $A \setminus D = \{x \in \mathbb{N} \mid x \text{ is even and not a perfect square}\}$
- f.) $D \setminus A = \{x \in \mathbb{N} \mid x \text{ is a perfect square and not even}\}$
- g.) $B \setminus C = x \text{ is odd and not prime}$
- h.) $C \setminus B = \{2\}$
- i.) $B \setminus D = x \text{ is odd and not a perfect square}\}$
- j.) $D \setminus B = x \text{ is a perfect square and is even}$
- k.) $C \setminus D = C$
- f.) $D \setminus C = D$
- g.) $A \setminus A = B \setminus B = C \setminus C = D \setminus D = \emptyset$

□

Theorem 1.13. Let A, B and X be sets. Then:

- a) $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$
- b) $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$

Proof of (a). We prove this theorem through double inclusion.

Firstly, let $a \in X \setminus (A \cup B)$, where a is some arbitrary element in the set. By **Definition 1.11**, we therefore can further state that

$$a \in X, a \notin A \cup B$$

$$a \in X, a \notin A, a \notin B$$

Therefore, we state that a is in X , but not in A nor in B .

$$a \in (X \setminus A) \cap (X \setminus B)$$

As $a \in X \setminus (A \cup B)$, $a \in (X \setminus A) \cap (X \setminus B)$, and a is some arbitrary element, we can state that:

$$X \setminus (A \cup B) \subset (X \setminus A) \cap (X \setminus B)$$

Secondly, let $b \in (X \setminus A) \cap (X \setminus B)$, where b is some arbitrary element. When expanded, we can state that

$$b \in X \setminus A \text{ and } b \in X \setminus B$$

As b is not in either in Set A nor in Set B , we can conclude that

$$b \in X, b \notin A, b \notin B$$

This can be written as

$$b \in X \text{ and } b \notin (A \cup B)$$

This may alternatively be written as

$$b \in X \setminus (A \cup B)$$

Therefore, we can state that b is an element in X , not in Set A , and not in set B . As $b \in X, b \notin A, b \notin B$, we may write this as

$$(X \setminus A) \cap (X \setminus B) \subset X \setminus (A \cup B)$$

As a result, we have both expressions that are subsets of one another.

$$X \setminus (A \cup B) \subset (X \setminus A) \cap (X \setminus B)$$

$$(X \setminus A) \cap (X \setminus B) \subset X \setminus (A \cup B)$$

In conclusion, as both expressions are subsets of one another, we can state that the two expressions are equal. As a result, we have proven the theorem.

$$X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$$

□

Proof of (b). We prove this theorem through double inclusion

Firstly, let $a \in X \setminus (A \cap B)$, where a is some arbitrary element in the set. This implies that $a \in X$, and $a \notin A \cap B$. Therefore, we can go further and say that $a \notin (A \cup B)$ and $a \in X$.

Therefore, we get the statement that

$$(a \in X \text{ and } a \notin A) \text{ or } (a \in X \text{ and } a \notin B)$$

We can write this as

$$a \in X \setminus A \text{ or } a \in X \setminus B$$

$$(X \setminus A) \cup (X \setminus B)$$

As $a \in X \setminus (A \cap B)$, we can state that

$$X \setminus (A \cap B) \in (X \setminus A) \cup (X \setminus B)$$

Secondly, suppose $a \in (X \setminus A) \cup (X \setminus B)$, where a is an arbitrary element in the set.

When we expand the expression, we state that

$$(a \in X \text{ and } a \notin A) \text{ or } (a \in X \text{ and } a \notin B)$$

As $a \notin A$ and $a \notin B$ we can say that $a \notin (A \cup B)$. Therefore, as we can state that

$$a \in X \setminus (A \cap B)$$

Hence, as $a \in X \setminus (A \cap B)$, $a \in (X \setminus A) \cup (X \setminus B)$, and a is an arbitrary element we can state that

$$(X \setminus A) \cup (X \setminus B) \in X \setminus (A \cap B)$$

Therefore through double inclusion, as $(X \setminus A) \cup (X \setminus B) \in X \setminus (A \cap B)$ and $X \setminus (A \cap B) \in (X \setminus A) \cup (X \setminus B)$, we can state that

$$X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$$

Sometimes we will encounter families of sets. The definitions of intersection/union can be extended to infinitely many sets.

Definition 1.14. Let $\mathcal{A} = \{A_\lambda \mid \lambda \in I\}$ be a collection of sets indexed by a nonempty set I . Then the intersection and union of $\mathcal{A}$ are the sets

$$\bigcap_{\lambda \in I} A_\lambda = \{x \mid x \in A_\lambda, \text{ for all } \lambda \in I\},$$

and

$$\bigcup_{\lambda \in I} A_\lambda = \{x \mid x \in A_\lambda, \text{ for some } \lambda \in I\}.$$

Theorem 1.15. Let X be a set, and let $\mathcal{A} = \{A_\lambda \mid \lambda \in I\}$ be a nonempty collection of sets. Then:

$$1. X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right) = \bigcap_{\lambda \in I} (X \setminus A_\lambda)$$

$$2. X \setminus \left(\bigcap_{\lambda \in I} A_\lambda \right) = \bigcup_{\lambda \in I} (X \setminus A_\lambda).$$

Proof of (1). We prove this theorem through double inclusion.

Firstly, suppose $a \in X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right)$. By **Definition 1.14** and **Definition 1.11**, we may say that a is an element where $a \in X$ and $a \notin \bigcup_{\lambda \in I} A_\lambda$. We may alternatively write this as $a \notin A_\lambda$ for all $\lambda \in I$. Therefore, we make the claim that

$$a \in X \text{ and } \forall \lambda \in I, a \in (X \setminus A_\lambda)$$

This can alternatively be written as

$$a \in \bigcap_{\lambda \in I} (X \setminus A_\lambda)$$

As $a \in X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right)$ and $a \in \bigcap_{\lambda \in I} (X \setminus A_\lambda)$, and a is an arbitrary element in both sets, we may write

$$X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right) \subset \bigcap_{\lambda \in I} (X \setminus A_\lambda)$$

Secondly, suppose $a \in \bigcap_{\lambda \in I} (X \setminus A_\lambda)$. By **Definition 1.14** and **Definition 1.11**, we can alternatively state that for all sets $A_\lambda, \forall \lambda \in I, a \in X \setminus A_\lambda$. Therefore we can say that a is present in X and not present in any A_λ for all λ in I . We may write this as

$$a \in X \text{ and } a \notin A_\lambda, \forall \lambda \in I,$$

As we state that $a \notin A_\lambda$ for all λ in I , we can alternatively say that a is not an element in the union set of all A_λ . We may write this as

$$a \notin \left(\bigcup_{\lambda \in I} A_\lambda \right)$$

As $a \in X$ and $a \in \bigcap_{\lambda \in I} (X \setminus A_\lambda)$, we may write this alternatively as

$$\bigcap_{\lambda \in I} (X \setminus A_\lambda) \subset X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right)$$

Finally, as both expressions are sets of one another, we may conclude at the statement that

$$X \setminus \left(\bigcup_{\lambda \in I} A_\lambda \right) = \bigcap_{\lambda \in I} (X \setminus A_\lambda)$$

□

Proof of (2). We prove this theorem through double inclusion

Firstly, let $a \in X$

By **Definition 1.14** and **Definition 1.11**, we can say that a is not present in any of the elements in any A_λ

$$a \notin (\cap_{\lambda \in I} A_\lambda)$$

Therefore, for $\forall \lambda \in I, a \notin A_\lambda$. Hence, we can state that the common element missing in A_λ is a .

This may be written as

$$a \in X, a \notin \cup_{\lambda \in I} (A_\lambda)$$

$$a \in X \setminus \cup_{\lambda \in I} (A_\lambda)$$

As $a \in X \setminus (\cap_{\lambda \in I} A_\lambda)$ and $a \in X \setminus \cup_{\lambda \in I} (A_\lambda)$, we may write this as

$$X \setminus \left(\bigcap_{\lambda \in I} A_\lambda \right) \subset X \setminus \cup_{\lambda \in I} (A_\lambda)$$

Secondly, let $a \in \cup_{\lambda \in I} (X \setminus A_\lambda)$.

We may also write this as

$$(a \in X) \text{ and } (a \notin \bigcap_{\lambda \in I} A_\lambda)$$

This can alternatively be written as

$$a \in X \setminus \bigcap_{\lambda \in I} A_\lambda$$

As $a \in \cup_{\lambda \in I} (X \setminus A_\lambda)$, we can therefore state that

$$\bigcup_{\lambda \in I} (X \setminus A_\lambda) \subset X \setminus \left(\bigcap_{\lambda \in I} A_\lambda \right)$$

Finally, as both expressions are sets of one another, we may conclude at the statement that

$$\bigcup_{\lambda \in I} (X \setminus A_\lambda) = X \setminus \left(\bigcap_{\lambda \in I} A_\lambda \right)$$

□

Definition 1.16. Let A and B be two nonempty sets. The *Cartesian product* of A and B is the set of ordered pairs

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

If (a, b) and $(a', b') \in A \times B$, we say that (a, b) and (a', b') are *equal* if and only if $a = a'$ and $b = b'$. In this case, we write $(a, b) = (a', b')$.

Functions

Definition 1.17. Let A and B be two nonempty sets. A *function* f from A to B is a subset $f \subset A \times B$ such that for all $a \in A$ there exists a unique $b \in B$ satisfying $(a, b) \in f$. To express the idea that $(a, b) \in f$, we most often write $f(a) = b$. To express that f is a function from A to B in symbols we write $f: A \rightarrow B$.

Exercise 1.18. Let the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(n) = 2n$. Write f as a subset of $\mathbb{Z} \times \mathbb{Z}$.

Answer. Let $f(n) = 2n$

We can write the subset of Cartesian Products of f as

$$f = \{a, f(a) \mid a \in \text{domain}\}$$

The subset of $f \subset \mathbb{Z} \times \mathbb{Z}$ would henceforth be written as

$$f = \{(n, 2n) \mid n \in \mathbb{Z}\}$$

□

Definition 1.19. Let $f: A \rightarrow B$ be a function. The *domain* of f is A and the *codomain* of f is B . If $X \subset A$, then the *image of X under f* is the set

$$f(X) = \{f(x) \in B \mid x \in X\}.$$

If $Y \subset B$, then the *preimage of Y under f* is the set

$$f^{-1}(Y) = \{a \in A \mid f(a) \in Y\}.$$

Exercise 1.20. Must $f(f^{-1}(Y)) = Y$ and $f^{-1}(f(X)) = X$? For each, either prove that it always holds or give a counterexample.

Proof. Let $f: A \rightarrow B$ in which the function is defined as $f(n) = n$. Let A and B be two sets where $A = \{1, 2\}$ and $B = \{1, 2, 3\}$.

We want to show there is a counterexample to the statement that $f(f^{-1}(Y)) = Y$ must be true. Suppose we have sets X and Y where $X = A$ and $Y = \{2, 3\}$. By **Definition 1.19**, we may say that

$$f^{-1}(Y) = \{a \in A \mid f(a) \in Y\} = \{2\}$$

Then, we may say that

$$f(f^{-1}(Y)) = f(\{2\}) = \{2\}$$

As this does not equal Y , this provides a counterexample to the statement that $f(f^{-1}(Y)) = Y$ must be true.

Secondly, let $A = \{1, 2, 3\}$ and $B = \{1, 2\}$ and let $f: A \rightarrow B$ where $f(1) = 1$, $f(2) = 2$, and $f(3) = 2$.

We want to prove the statement that $f^{-1}(f(X)) = X$ must be true. Suppose $X \subset A = \{2\}$, then $f(X) = 2$. We may therefore state that

$$f^{-1}(f(X)) = f^{-1}(\{2\}) = \{x \in A : f(x) = 2\} = \{2, 3\}$$

Therefore, as $f^{-1}(f(X)) = \{2, 3\}$ we may state that $f^{-1}(f(X)) \neq X$. Hence, this works as a counterexample to the statement that $f^{-1}(f(X)) = X$ must be true. □

Definition 1.21. A function $f: A \rightarrow B$ is *surjective* (also known as ‘onto’) if, for every $b \in B$, there is some $a \in A$ such that $f(a) = b$. The function f is *injective* (also known as ‘one-to-one’) if for all $a, a' \in A$, if $f(a) = f(a')$, then $a = a'$. The function f is *bijective*, (also known as a bijection or a ‘one-to-one’ correspondence) if it is surjective and injective.

Exercise 1.22. Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = n + 2$. Is f injective? Is f surjective?

Proof. We want to prove $f: \mathbb{N} \rightarrow \mathbb{N}$ is injective. Suppose b, b' and arbitrary elements in $\mathbb{N}$. By **Definition 1.21**, we can say that if $f(b) = f(b')$, then we may say that $b = b'$. Therefore, we may use the following to prove injectivity.

$$f(b) = b + 2 \text{ and } f(b') = b' + 2$$

$$b + 2 = b' + 2$$

$$b = b'$$

As we have taken arbitrary values for b , we may say that the function is injective.

Secondly, we want to prove $f: \mathbb{N} \rightarrow \mathbb{N}$ is not surjective. By **Definition 1.21**, we may disprove surjection by stating that for every $b \in B$, there is $a \notin A$ such that $f(a) = b$.

Let $b = 1$

$$a + 2 = b$$

$$a + 2 = 1$$

$$a + 2 = 1$$

$$a = 1 - 2 = -1$$

$$a = -1$$

This is the only possible value a that satisfies $a + 2 = 1$ because the equation has only one possible solution, as it is linear and may not contain multiple solutions. Therefore, the only possible solution for a is for $f: \mathbb{N} \rightarrow \mathbb{N}$.

As $a \notin \mathbb{N}$, not every $b \in B$, the function is not surjective. □

Exercise 1.23. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = x + 2$. Is f injective? Is f surjective?

Proof. We want to prove $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is injective.

Suppose b, b' and arbitrary elements in $\mathbb{Z}$. By **Definition 1.21**, a function is injective if for any two elements, b and b' , in the domain $\mathbb{Z}$, if $f(b) = f(b')$, then it must follow that $b = b'$. Therefore, if $b = b'$ when $f(b) = f(b')$ we may say that the function is injective.

Let b and b' be arbitrary elements in the domain $\mathbb{Z}$.

$$\begin{aligned}f(b) &= b + 2 = f(b') = b' + 2 \\b + 2 &= b' + 2 \\b &= b'\end{aligned}$$

Hence, we can state that the function is injective.

Secondly, we can prove the surjection through a proof by construction. The domain of $\mathbb{Z}$ is an infinite set that contains all positive and negative integers. Let $y \in \mathbb{Z}$. By the domain of $\mathbb{Z}$, we can say that $y - 2 \in \mathbb{Z}$ by closure.

, when the image $y - 2$ is used as the pre-image, we get the original value of the image

$$f(y - 2) = (y - 2) + 2 = y$$

Therefore, we state that f is a subjective function.

□

Exercise 1.24. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = n^2$. Is f injective? Is f surjective?

Proof. We want to prove $f : \mathbb{N} \rightarrow \mathbb{N}$ is injective. Suppose a, b are elements in $\mathbb{N}$. By **Definition 1.21**, a function is injective if for any two elements, a and b , in the domain $\mathbb{N}$, if $f(a) = f(b)$, then it must follow that $a = b$. Therefore, if $a = b$ when $f(a) = f(b)$, we may say that the function is injective.

Let $a, b \in \mathbb{N}$. Therefore $f(a) = a^2$ and $f(b) = b^2$. If two numbers are different, the square of one number cannot equal another. If $a \neq b$ then $a^2 \neq b^2$ and therefore $f(a) \neq f(b)$. Hence, as we have established that the domain is $\mathbb{N}$, the only way in such the square of a number may be the same is if they are the same the number.

$$\begin{aligned}f(a) &= f(b) \\a^2 &= b^2\end{aligned}$$

$a = b$ only if a and b are the same element

Hence we can state that $f(a) = f(b)$ only if $a = b$. Therefore, we may prove injectivity.

Secondly, we can prove that the function is not surjective through contradiction. Consider the element $y = 3$ in the codomain $\mathbb{N}$. For f to be surjective, there must exist some $b \in \mathbb{N}$ such that

$$f(b) = 3.$$

Let $b \in \mathbb{N}$. The set $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ and b must be an element in this set that satisfies $b^2 = 3$. To understand if this is possible we may go through the elements in $\mathbb{N}$. If $b = 1$, then $1^2 = 1 \neq 3$. Therefore, we know $b \neq 1$. If $b = 2$, then $2^2 = 4 \neq 3$. Therefore, we know $b \neq 2$. By the existence of multiplicative identity, we observe that if $b > 1$, $b^2 > b \cdot 1$ as $b \cdot 1 = b$. Therefore, $b \not> 2$ as if $b = 2$ then $b > 3$. Hence, if $b^2 = 3$, $b > 1$ and $b < 2$. Therefore $b^2 \neq 3$ and therefore the function is not injective. □

Exercise 1.25. Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = x^2$. Is f injective? Is f surjective?

Proof of 1.25. We can prove that the function is not injective through counterexample. By **Definition 1.21**, we say a function is injective if for an arbitrary $a, a' \in \mathbb{Z}$, if $f(a) = f(a')$, then $a = a'$. To disprove this, one of the conditions would need to be false.

Let $a, a' \in \mathbb{Z}$, where $a = 2, a' = -2$.

$$\begin{aligned} f(2) &= f(-2) \\ 2^2 &= (-2)^2 \\ 4 &= 4 \end{aligned}$$

However, as $2 \neq -2$ we can therefore conclude that function f is not injective.

We prove that the function is not surjective through counterexample. By **Definition 1.21** we say a function is surjective if $\forall b \in B$ there is some $a \in A$ such that $f(a) = b$.

Let $b \in \mathbb{Z}$, where $b = -4$.

$$\begin{aligned} f(a) &= -4 \\ x^2 &= -4 \\ x &= \sqrt{-4} \end{aligned}$$

The square root of -4 is an imaginary number, therefore we conclude is no $a \in \mathbb{Z}$. Hence, we say that f is not surjective. □

Definition 1.26. Let $f : A \rightarrow B$ and $g : B \rightarrow C$. Then the *composition* $g \circ f : A \rightarrow C$ is defined by $(g \circ f)(x) = g(f(x))$, for all $x \in A$.

Proposition 1.27. Let A, B , and C be sets and suppose that $f : A \rightarrow B$ and $g : B \rightarrow C$. Then $g \circ f : A \rightarrow C$ and

- a) if f and g are both injections, so is $g \circ f$.
- b) if f and g are both surjections, so is $g \circ f$.

c) if f and g are both bijections, so is $g \circ f$.

Proposition 1.28. Suppose that $f: A \rightarrow B$ is bijective. Then there exists a bijection $g: B \rightarrow A$ that satisfies $(g \circ f)(a) = a, \forall a \in A$, and $(f \circ g)(b) = b$, for all $b \in B$. The function g is often called the inverse of f and denoted f^{-1} . It should not be confused with the preimage.

Definition 1.29. We say that two sets A and B are in *bijective correspondence* when there exists a bijection from A to B or, equivalently, from B to A .

Cardinality

Definition 1.30. Let $n \in \mathbb{N}$ be a natural number. We define $[n]$ to be the set $\{1, 2, \dots, n\}$. Additionally, we define $[0] = \emptyset$.

Definition 1.31. A set A is *finite* if $A = \emptyset$ or if there exists a natural number n and a bijective correspondence between A and the set $[n]$. If A is not finite, we say that A is *infinite*.

Theorem 1.32. Let $n, m \in \mathbb{N}$ with $n < m$.

Then there does not exist an injective function $f: [m] \rightarrow [n]$.

Hint: Fix $k \in \mathbb{N}$. Prove, by induction on n , that for all $n \in \mathbb{N}$, there is no injective function $f: [n+k] \rightarrow [n]$.

Proof of 1.32. We want to prove that if $n < m$ then there does not exist an injective function $f: [m] \rightarrow [n]$. We want to prove this by contradiction.

Suppose there exist $n, m \in \mathbb{N}$ such that $n < m$. Then, there must exist some $k \in \mathbb{N}$ such that $n+k = m$. Therefore, we may write the function f as $f: [n+k] \rightarrow [n]$. We want to prove by induction that for all $n \in \mathbb{N}$ that there is no injective function $f: [n+k] \rightarrow [n]$.

Firstly, we want to prove the base case. Suppose that $n = 1$. We want to show that $f: [1+k] \rightarrow [1]$ is not injective.

If $n = 1$, then $k \geq 2$. Therefore, there exist at least two elements that correspond to the one element in the codomain. Hence, it is not necessary that $\forall k, k' \in \mathbb{N}$ if $f(k) = f(k')$ then $k = k'$. Thus, we have shown a lack of injectivity in the base case.

Now, let us assume that $f: [n+k] \rightarrow [n]$ must be true. We want to now show that there is no injective function $g: [(n+1)+k] \rightarrow [n+1]$. We can show this by contradiction.

Suppose that $g: [(n+1)+k] \rightarrow [n+1]$ is injective. Then we must prove two cases.

Case 1: $g: ((n+1)+k) = (n+1)$.

Now, we define a function S such that $S = [(n+1)+k]$. Then, we may write $|S| = (n+1)+k$. Now, assign the greatest element of S as x such that $x = n+1+k$. Now, we may define a function S' such that $S' = S \setminus \{x\}$. Therefore, we can say

$$|S'| = |S \setminus \{x\}| = n+k$$

Similarly, we define the codomain as $[n+1] = \{1, 2, \dots, n+1\}$. Let y be the greatest element of $[n+1]$ such that $y = n+1$. Now, we may define a function R' as

$$R' = [n+1] \setminus \{n+1\} = \{1, 2, \dots, n\}$$

Hence, $|R'| = n$.

If g is injective, then the restriction of g to D' must map each element of S' to a distinct element of R' .

$$g|_{D'} : D' \rightarrow C'.$$

However, $|D'| = n+k > n = |C'|$, so no injective function exists from D' to C' by the basic counting principle. This is a contradiction.

Hence, our assumption that $g(n+1+k) = n+1$ is not true.

Case 2: $g(n+1+k) \neq n+1$.

Since g has codomain $[n+1] = \{1, 2, \dots, n+1\}$ and is injective, there must exist some $y \in [(n+1)+k]$ such that $g(y) = n+1$. Because $g(n+1+k) \neq n+1$, we have $y \neq n+1+k$. Let

$$D' = [(n+1)+k] \setminus \{y\},$$

so that $|D'| = (n+1) + k - 1 = n+k$.

For any $x \in D'$, we have $g(x) \neq n+1$ (since $g(y) = n+1$ and g is injective). Therefore,

$$g(D') \subseteq [n+1] \setminus \{n+1\} = [n].$$

Thus g determines an injective function $g : D' \rightarrow [n]$. But $|D'| = n+k > n = |[n]|$, and hence no such injective function can exist. This is a contradiction.

Therefore Case 2 is impossible.

□

Theorem 1.33. *Let A be a finite set. Suppose that A is in bijective correspondence both with $[m]$ and with $[n]$. Then $m = n$.*

The preceding result allows us to make the following important definition.

Definition 1.34 (Cardinality of a finite set). If A is a finite set that is in bijective correspondence with $[n]$, then we say that the *cardinality* of A is n , and we write $|A| = n$. (By Theorem 1.33, there is exactly one such natural number n .) We define the cardinality of the empty set to be 0.

Exercise 1.35. Let A and B be finite sets.

- a) If $A \subset B$, then $|A| \leq |B|$.
- b) Let $A \cap B = \emptyset$. Then $|A \cup B| = |A| + |B|$.
- c) $|A \cup B| + |A \cap B| = |A| + |B|$
- d) $|A \times B| = |A| \cdot |B|$.

Important Note: In future scripts you may assume basic properties of finite sets and methods of counting elements without having to refer back to the notions presented in this script. When $|A| = n$, we say that A contains n elements.

Definition 1.36. A set A is said to be *countable* either if it is finite or if it is in bijective correspondence with $\mathbb{N}$. An infinite set that is not countable is called *uncountable*.

Exercise 1.37. Prove that $\mathbb{Z}$ is a countable set.

Theorem 1.38. Every subset of $\mathbb{N}$ is countable.

Hint: when A is an infinite subset of $\mathbb{N}$, construct a bijection $f: \mathbb{N} \rightarrow A$ inductively/ recursively. This looks similar to a proof by induction: define an initial term (or terms) explicitly and then present a rule that defines $f(n+1)$ assuming that $f(1), \dots, f(n)$ have already been defined. For example, the factorial of n is defined inductively by letting $0! = 1$ and

$$(n+1)! = (n+1) \cdot n!$$

for $n \geq 0$. After constructing your function you must verify that it is indeed a bijection.

Proof of 1.38. We want to prove that a subset of $\mathbb{N}$ is countable, regardless of whether the subset is finite or infinite.

Firstly, suppose the subset $A \subset \mathbb{N}$ is finite. If the set is finite, by **Definition 1.36** we know that the set is countable. Therefore, a finite subset of $\mathbb{N}$ is countable.

Secondly, suppose the subset $B \subset \mathbb{N}$ is infinite. Let $f: \mathbb{N} \rightarrow B$ and let $f(n) \in B$. As B is infinite, we may say that $B' = B \setminus f(n)$ and that $B' \neq \emptyset$. Therefore, there exists an element $f(n+1)$ such that $f(n+1) \in B'$ and $f(n+1) \neq f(n)$.

Through this, we know that all elements of set B are distinct. Therefore, for all $b, b' \in B$, if $f(b) = f(b')$ then $b = b'$ as no value repeats itself. Hence, we may say the function is injective.

To prove surjectivity, let $b \in B$. Suppose $B_n = B \setminus \{f(1), f(2), \dots, f(n)\}$ where B_n represents the elements that have already been selected. Let $f(n+1) \in B$ and $f(n+1) \notin B_n$. Suppose $b \notin B_n$. In this case, there are two possible cases:

Case 1: $f(n+1) = b$, in which case we have found the element, or

Case 2: $f(n+1) \neq b$, in which case only $f(n+1)$ is removed from B_n . As B is infinite, we may constantly find another value for n such that $f(n) \in B$. Eventually, as one new element of B is removed at each stage and we are performing infinitely many steps as B is infinite, every element of B is eventually chosen. Out of all elements, one $f(n) = b$.

Therefore, there exists $n \in \mathbb{N}$ such that $f(n) = b$, so f is surjective.

As we have proved both surjectivity and injectivity, we may say that there is a bijective correspondence with $\mathbb{N}$. Therefore, we have proven that all subsets of $\mathbb{N}$ are countable. $\square$

Theorem 1.39. *If there exists an injection $f : A \rightarrow B$ where B is countable, then A is countable.*
Hint: Use Theorem 1.38.

Corollary 1.40. *Every subset of a countable set is also countable.*

Corollary 1.41. *If there exists a surjection $g : B \rightarrow A$ where B is countable, then A is countable.*
Hint: Use Theorem 1.39.

Exercise 1.42. Prove that $\mathbb{N} \times \mathbb{N}$ is countable by considering the function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ given by $f(n, m) = (10^n - 1)10^m$.

(Alternatively you could use either one of the functions $g(n, m) = 2^n \cdot 3^m$ and $h(n, m) = \binom{n+m}{2} + n$.)

Proof. We want to show that $\mathbb{N} \times \mathbb{N}$ is countable.

We may say the function f is injective as if $(n, m) \neq (n', m')$, then $f(n, m) \neq f(n', m')$, as the numbers produced by the function have a unique pattern of n nines followed by m zeros.

Then, let the image of f be defined as

$$f(\mathbb{N} \times \mathbb{N}) = \{f(n, m) : n, m \in \mathbb{N}\}.$$

By its construction, every element of $f(\mathbb{N} \times \mathbb{N})$ is a natural number. Therefore, we may say the function $f(\mathbb{N} \times \mathbb{N})$ is a subset of $\mathbb{N}$. Since we already know that every subset of $\mathbb{N}$ is countable, it follows that $f(\mathbb{N} \times \mathbb{N})$ is countable.

Finally, because f is injective, $\mathbb{N} \times \mathbb{N}$ is in one-to-one correspondence with $f(\mathbb{N} \times \mathbb{N})$. Therefore, $\mathbb{N} \times \mathbb{N}$ is countable. $\square$

Additional Exercises

In all exercises you are expected to prove your answer, unless explicitly stated otherwise.

1. In each of the following, write out the elements of the sets.

a) $(\{n \in \mathbb{Z} \mid n \text{ is divisible by } 2\} \cap \mathbb{N}) \cup \{-5\}$

Proof. Let $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, $\mathbb{N} = \{0, 1, 2, \dots\}$

As $\mathbb{Z} \cap \mathbb{N} = \mathbb{N}$, we can state that $\{n \in \mathbb{N} \mid n \text{ is divisible by } 2\}$.

Therefore, we write out the elements of the set as follows

$$n \in \{2, 4, 6, \dots, 2n\} \cup \{-5\}$$

b) $\{F, G, H\} \times \{5, 8, 9\}$

Proof. The Cartesian product of a set is the product of the two sets as ordered pairs. The Cartesian product of $\{F, G, H\} \times \{5, 8, 9\}$ would be

$$\{(F, 5), (F, 8), (F, 9), (G, 5), (G, 8), (G, 9), (H, 5), (H, 8), (H, 9)\}$$

$\square$

c) $\{[n] \mid n \in \mathbb{N}, 1 \leq n \leq 3\}$

Proof. The cardinality of $[n]$ within the domain $1 \leq n \leq 3$. Therefore, we can state that the elements in the set would be

$$\{\{1\}, \{1, 2\}, \{1, 2, 3\}\}$$

□

d) $\{(x, y) \in \mathbb{N} \times \mathbb{N} \mid y = 2x, x = 2y\}$

Proof. Let $y = 2x$ and $x = 2y$. To find the Cartesian product, we can substitute the first equation into the second course.

$$x = 2y$$

$$x = (2)2x = 4x$$

$$x - x = 4x - x$$

$$0 = 3x$$

$$x = 0$$

Now, substitute $x = 0$ back into the first equation to find y

$$y = 2x$$

$$y = 2(0)$$

$$y = 0$$

The only ordered pair (x, y) that satisfies the equations $y = 2x$ and $x = 2y$ is $(0, 0)$. However, this is not an element within the domain and therefore there is no value in the set

$$\{(x, y) \in \mathbb{N} \times \mathbb{N} \mid y = 2x, x = 2y\} = \emptyset$$

□

e) $(\{1, 2\} \times \{1, 2\}) \times \{1, 2\}$

Proof. The Cartesian product of two sets can be defined as

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Let $A = (1, 2), (1, 2)$ and $B = 1, 2$. As the elements of the Cartesian product are all ordered pairs (a, b) where $a \in A$ and $b \in B$

Therefore, we can write the Cartesian Product as

$$(\{1, 2\} \times \{1, 2\}) \times \{1, 2\} = \{((1, 1), 1), ((1, 1), 2), ((1, 2), 1), ((1, 2), 2), ((2, 1), 1), ((2, 1), 2), ((2, 2), 1), ((2, 2), 2)\}$$

□

f) $(\{1, 2\} \cup \{1, 2\}) \cup \{1, 2\}$

Proof. The union of a set is defined as the collection of elements in either set or both.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Therefore, we can say that if $a \in A$, or $b \in B$, or both, then the element is part of the union set.

Let $A = \{1, 2\} \cup \{1, 2\}$ and $B = \{1, 2\}$. As we are finding the composites of both sets, $\{1, 2\} \cup \{1, 2\} = \{1, 2\}$

$$\begin{aligned} &(\{1, 2\} \cup \{1, 2\}) \cup \{1, 2\} \\ &\{1, 2\} \cup \{1, 2\} \\ &\{1, 2\} \end{aligned}$$

Hence, we can say that $(\{1, 2\} \cup \{1, 2\}) \cup \{1, 2\} = \{1, 2\}$.

□

g) $\{1, 2\} \cup \emptyset$

Proof. A set is a union of two sets when every element present in Set A, Set B, or both sets is present. Let Set A = $\{1, 2\}$ and Set B = $\emptyset$. As the empty set contains no elements, we may state that $\{1, 2\} \cup \emptyset = \{1, 2\}$.

□

h) $\{1, 2\} \cap \emptyset$

Proof. A set is an intersection of two sets when every element in Set A is present in Set B. Let Set A = $\{1, 2\}$ and Set B = $\emptyset$. The empty set contains no values, and therefore none of the values in Set A are present in Set B. Hence we can conclude that $\{1, 2\} \cap \emptyset = \emptyset$.

□

i) $\{1, 2\} \cup \{\emptyset\}$

Proof. Let Set A = $\{1, 2\}$ and Set B = $\{\emptyset\}$. The union set of Set A and B would contain the elements of Set A and Set B. In this case, as $\emptyset$ is an element within the set, and not the value of the set itself, we state that the $\{1, 2\} \cup \{\emptyset\} = \{1, 2, \emptyset\}$

□

j) $\{1, 2\} \cap \{\emptyset\}$

Proof. Let Set A = $\{1, 2\}$ and Set B = $\{\emptyset\}$. The intersection set of Set A and Set B would require all elements in Set A to be in Set B. Therefore, we can say that $\{1, 2\} \cap \{\emptyset\} = \emptyset$ as none of the elements in Set A are in Set B.

□

k) $\{\{a\} \cup \{b\} \mid a \in \mathbb{N}, b \in \mathbb{N}, 1 \leq a \leq 4, 3 \leq b \leq 5\}$

Proof. Let Set A and Set B be two individual sets, where set $a \in \{1, 2, 3, 4\}$ and $b \in \{3, 4, 5\}$. The union set of these two sets would ensure every element is in Set A, Set B, or in both Set A and Set B. Hence,

$$\{a\} \cup \{b\} \mid a \in \mathbb{N}, b \in \mathbb{N}, 1 \leq a \leq 4, 3 \leq b \leq 5 = \\ \{\{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4\}, \{3\}\}$$

□

1) $\{\{12\}\} \cup \{12\}$

Proof. The set $\{\{12\}\}$ contains the sets $\{12\}$. The set $\{12\}$ contains the element 12. The union of the two sets contains the elements that are in both sets. Therefore $\{\{12\}\} \cup \{12\} = \{\{12\}, 12\}$. □

2. Let A and B be subsets of the set X . The symmetric sum $A \oplus B$ (sometimes also called symmetric difference) of sets A and B is defined by

$$A \oplus B = (A \setminus B) \cup (B \setminus A).$$

Prove that

$$A \oplus B = (A \cup B) \cap [X \setminus (A \cap B)].$$

Proof. We want to prove that $A \oplus B = (A \setminus B) \cup (B \setminus A)$ may be written as $A \oplus B = (A \cup B) \cap [X \setminus (A \cap B)]$. We will prove this by double inclusion.

Suppose $x \in A \oplus B$, by the above stated definition we can state that, $x \in (A \setminus B) \cup (B \setminus A)$. This means that $x \in A \setminus B$ or $x \in B \setminus A$. Therefore, x is either exclusive to A or exclusive to B . As a result, $x \in A \cup B$ and $x \in X \setminus (A \cap B)$. Therefore, we can say that $x \in (A \cup B) \cap [X \setminus (A \cap B)]$.

As $x \in (A \setminus B) \cup (B \setminus A)$, we can state that $A \oplus B \subset (A \cup B) \cap [X \setminus (A \cap B)]$.

Secondly, let us assume $x \in (A \cup B) \cap [X \setminus (A \cap B)]$. This means that $x \in A \cup B$ and $x \in X \setminus (A \cap B)$. As $x \in A \cup B$ we can say $x \in A$, $x \in B$, or $x \in A \cap B$. Furthermore, as $x \in X \setminus (A \cap B)$ we can say $x \notin A \cap B$. Therefore, x is either exclusive to A or exclusive to B . An element x must be only in A or only in B . As a result, we can say that one of the following is true

$$x \in A \text{ and } x \notin B, \text{ and therefore } x \in A \setminus B$$

$$x \in B \text{ and } x \notin A, \text{ and therefore } x \in B \setminus A$$

.

We may write these two equations as $x \in (A \setminus B) \cup (B \setminus A)$. The expression $x \in (A \setminus B) \cup (B \setminus A)$ may also be written as $A \oplus B$. Therefore, we may write that $[(A \setminus B) \cup (B \setminus A)] \subset$

$$[(A \setminus B) \cup (B \setminus A)].$$

As both expressions are subsets of one another, we can say that the equations are equal, and thereby proving that $A \oplus B = (A \cup B) \cap [X \setminus (A \cap B)]$ is valid.

□

3.

Definition 1.43. Let A be a set. The “power set” of A , denoted $\wp(A)$, is the set of all subsets of A ; that is, $\wp(A) = \{B \mid B \subset A\}$.

- (a) If A is a set, show that $\wp(A) \neq \emptyset$.
- (b) Let $\emptyset$ be the empty set. Write down the elements of $\wp(\wp(\emptyset))$.

Proof of i. If A is any set, we know that $\emptyset \subset A$. Hence, $\emptyset \in \wp(A)$. Therefore, we can conclude that $\wp(A) \neq \emptyset$.

□

Proof of ii. As the empty set is put within the function again, we may say that $\wp(\wp(\emptyset)) = \{\emptyset, \{\emptyset\}\}$.

□

- (a) Let A, B, C be subsets of $\mathbb{N}$. Extend Theorem 1.7 by showing that, for any $k \in \mathbb{N}$

$$\begin{aligned} A &\subset A \cup B \cup C, \\ A \cap B \cap C &\subset A. \end{aligned}$$

Can this be extended to four sets A, B, C, D ? What about five? Is there any limit?

2. (a) Set $A = \{1, 2\}$, $B = \{3, 4\}$, and $f = \{(a, b) \mid a \in A, b \in B\}$.
Write out the elements of f . Is f a function $A \rightarrow B$?
- (b) Let $C = \{(1, 2), (2, 2), (3, 2)\}$.
Can C be a function? (For starters, what would A and B be?)
- (c) Write out the elements of the set $D = \{(b, a) \mid (a, b) \in C\}$, where C is as given in b).
Can D be a function?
3. Take the sets $A = \{1, 2, 3\}$ and $B = \{1, 4, 9\}$. Consider the following four statements:
 - (a) For all $a \in A$, there is some $b \in B$ such that $a^2 = b$.
 - (b) There is some $b \in B$ such that, for all $a \in A$, $a^2 = b$.
 - (c) There is some $b \in B$ such that $a^2 = b$ for all $a \in A$.
 - (d) For all $a \in A$, $a^2 = b$ for some $b \in B$.

Each statement is equivalent to exactly one other in the list. Which statements are true? Which pairs are equivalent to each other?

4. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be given by $f(n) = n^3$.
 - (a) Is f surjective? Is f injective?
 - (b) Let $A \subset \mathbb{N}$ be the set $\{1, 2, \dots, 30\}$. What is $f^{-1}(f(A))$? What is $f(f^{-1}(A))$?
5. Define $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(m, n) = mn$. Is f injective? Surjective? If $A \subset \mathbb{Z}$ is the set of even integers, what is $f^{-1}(A)$?
6. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.
 - (a) Suppose that f and $g \circ f$ are injective. Is g necessarily injective?

Proof of part(a). Suppose that f and $g(f(x))$ are injective functions. We want to show that g is not necessarily injective. We will prove this through counterexample.

Let $A = \{1, 2\}$, $B = \{a, b, c\}$, and $C = \{x, y\}$. We have that there exists two functions $f : A \rightarrow B$ and $g : B \rightarrow C$. Then, for function f let $f(1) = a$ and $f(2) = b$. For function g let $g(a) = x$, $g(b) = y$, and $g(c) = x$. By the definition of the functions we know that f is an injective function and g is not an injective function.

Now, we want to show that $g \circ f$ remains injective. We may write $g \circ f$ as $g(f(x)) : A \xrightarrow{f} B \xrightarrow{g} C$. Hence, by the definition of the function, we know that $\forall a, a' \in A$ if $f(a) = f(a')$ then $a = a'$. Thus, we have a counterexample to prove that g is not necessarily injective. $\square$

- (b) Suppose that g and $g \circ f$ are injective. Is f necessarily injective?

Proof of part (b). Suppose that g and $g(f(x))$ are injective functions. We want to show that f must be injective. We will prove this through contradiction.

Suppose f is not injective. Now, we may write $g \circ f$ as $g(f(x)) : A \rightarrow C$. As g is injective, we know that if $g(b) = g(b')$ then $b = b'$. Therefore we may say that

$$\text{if } g(f(a)) = g(f(a')) \text{ then we know that } f(a) = f(a')$$

As f is not injective, we may say that

$$\text{if } f(a) = f(a') \text{ it may be that } a \neq a'$$

However, this contradicts the known statement that $g \circ f$. Therefore, the statement that f is not injective is false. Thus, this works to prove that it is necessary for f to be injective. $\square$

- (c) Suppose that f and $g \circ f$ are surjective. Is g necessarily surjective?

Proof of part(c). Suppose that f and $g(f(x))$ are surjective functions. We want to show that g must be surjective. We will prove this through contradiction.

Suppose that g is not surjective. As f and $g \circ f$ are surjective, we know that $\forall b \in B, \exists a \in A$ such that $f(a) = b$ and $\forall c \in C, \exists a \in A$ such that $g \circ f(a) = c$.

As g is not surjective, we know that there exists some $c \in C$ such that for all $b \in B$ $g(b) \neq c$. We may write $g \circ f$ as $g \circ f : A \xrightarrow{f} B \xrightarrow{g} C$. We know that there exists some $a \in A$ such that $(g \circ f)(a) = c$ and that $f(a) = b$. Therefore, it must be that $g(b) = c$. This contradicts the original statement that g may not be surjective. Thus, this works to prove that it is necessary for g to be surjective.

□

- (d) Suppose that g and $g \circ f$ are surjective. Is f necessarily surjective?

Proof of Part(d). Suppose that g and $g(f(x))$ are surjective functions. We want to show that f is not necessarily surjective. We will prove this through counterexample.

Let $A = \{1\}$, $B = \{a, b\}$, and $C = \{x\}$. For the function f let $f(1) = a$ and $g(a) = g(b) = x$. Then, by the definition of f and g we know that g is surjective but f is not surjective.

Now we need to show that $g \circ f$ is surjective. We may write $g \circ f$ as $g \circ f : A \rightarrow C$. Therefore if $g \circ f$ is surjective we know that for all $c \in C, \exists a \in A$ such that $g(f(a)) = c$. As $x = g(a) = f(1)$ we know that $g \circ f$ is surjective. Thus, we have a counterexample that proves f must not necessarily be surjective for $g \circ f$ to be surjective. Thus, we have proved that f must not necessarily be surjective for $g \circ f$ to be surjective.

□

7. Let $f : A \rightarrow B$ be a function. Let $X \subset A$ and $Y \subset B$.

- (a) Prove that if f is surjective then $f(f^{-1}(Y)) = Y$.
- (b) Prove that if f is injective then $f^{-1}(f(X)) = X$.
- (c) Are the converse statements also true? i.e. If $f(f^{-1}(Y)) = Y$ for all subsets $Y \subset B$, must f be surjective? If $f^{-1}(f(X)) = X$ for all subsets $X \subset A$, must f be injective?

8. Let $f : A \rightarrow B$ be a bijection. Let g be the inverse function to f , given by Proposition 1.27. Let $Y \subset B$. Show that $g(Y) = f^{-1}(Y)$.

Note: $g(Y)$ denotes the image of Y under the map g and $f^{-1}(Y)$ denotes the preimage of Y under f . Thus when $g = f^{-1}$ exists as a function, the two possible interpretations of $f^{-1}(Y)$ coincide.

9. Recall the definition of power set, Definition 1.43.

- (a) Let A be any set. Show that there is no bijection between A and its power set $\wp(A)$. (Hint: If $f : A \rightarrow \wp(A)$ is any function, think about the set $B = \{a \in A \mid a \notin f(a)\} \subset A$.)

Proof. We want to prove that there exists no bijection between A and its power set $\wp(A)$. We will prove this through contradiction.

We want to show that there does not exist a surjection. Suppose there exists a bijection $f : A \rightarrow \wp(A)$.

Define the set B such that $B \subset A$ and is defined by $B = \{a \in A \mid a \notin f(a)\}$. As $B \subset A$, we may say that $B \in \wp(A)$.

Now, we know that if f is surjective, then for all $Y \in \wp(A)$, there exists some $a \in A$ such that $f(a) = Y$. To prove the contradiction, we want to show that there exists some $Y_0 \in \wp(A)$ such that for all $a \in A$, $f(a) \neq Y_0$.

Let $Y_0 = B$. Assume that there exists some $a_0 \in A$ such that $f(a_0) = B$. If $a_0 \in B$, then by definition $a_0 \notin f(a_0)$, hence $f(a_0) \neq B$. However, we know that $f(a_0) = B$ and therefore this is a contradiction. Then, if $a_0 \notin B$, then by definition $a_0 \in f(a_0)$, hence $f(a_0) \neq B$. Similarly, as we know that $f(a_0) = B$ this is a contradiction.

Thus, we can conclude that no such a_0 exists. Therefore, f is not surjective, hence it is not bijective. This completes the proof. $\square$

(b) Show that there is no injective map from $\wp(\mathbb{N})$ to $\mathbb{N}$.

Proof. We want to prove that there is no injective map from $\wp(\mathbb{N})$ to $\mathbb{N}$. We will prove this by contradiction.

Suppose, for contradiction, that such an injective function f exists.

Now, we may say that for each $n \in \mathbb{N}$, injectivity ensures that there is at most one subset $A \subseteq \mathbb{N}$ such that $f(A) = n$. Thus $f^{-1}(n)$ is well-defined whenever n lies in the image of f . Therefore we can define a set B as

$$B = \{n \in \mathbb{N} : n \notin f^{-1}(n)\}.$$

Since $B \subseteq \mathbb{N}$, we have $B \in \wp(\mathbb{N})$, so its image $f(B)$ is a natural number. Let the arbitrary element b be defined as $b = f(B)$.

We now consider whether b belongs to B .

Case 1: $b \in B$.

Suppose that $b \in B$. By the definition of B , we know that $b \notin f^{-1}(b)$. However, $f^{-1}(b) = B$ since $f(B) = b$. Hence $b \in B$ implies $b \notin B$, a contradiction.

Case 2: $b \notin B$

Suppose $b \notin B$. By the definition of B , we know that $b \in f^{-1}(b)$. Similarly, as $f^{-1}(b) = B$, this means $b \in B$. Hence, this contradicts our assumption that $b \notin B$.

In both cases we obtain a contradiction to the assumption. Thus, this works to prove that no such injective map $f : \wp(\mathbb{N}) \rightarrow \mathbb{N}$ can exist. $\square$

10. Let A be a set with cardinality n . Let $f : [n] \rightarrow A$ be a bijection. Show that $A = \{f(1), f(2), \dots, f(n)\}$ and deduce that we can write $A = \{a_1, a_2, \dots, a_n\}$.

Proof of 13. We want to prove that if $f : [n] \rightarrow A$ is a bijection, then $A = \{f(1), f(2), \dots, f(n)\}$ and it may be written as $A = \{a_1, a_2, \dots, a_n\}$.

Suppose, we have a function f such that $f : [n] \rightarrow A$ is a bijection. Therefore, we know that $\forall a \in A$ there exists a unique $x \in [n]$ such that $f(x) = a$. By **Definition 1.30**, we may write that $[n] = \{1, 2, \dots, n\}$. Thus, every element of A can be written in the form of $f(x)$ where $x \in [n]$. Therefore, we may write $f([n]) = \{f(1), f(2), \dots, f(n)\}$ and $A = f([n])$.

Now, we may write A as $A = \{f(1), f(2), \dots, f(n)\}$. Since f is injective, the values $f(1), f(2), \dots, f(n)$ are all distinct. Furthermore, we know that the cardinality of A and $[n]$ are the same there exists a bijection. Therefore we may enumerate the elements $f(1), f(2), \dots, f(n)$ to relabel them as $a_1, a_2, \dots, a_n$. Thus, we may conclude $A = \{a_1, a_2, \dots, a_n\}$. $\square$

11. Let $f : A \rightarrow B$ be a function.

- (a) Let X and Y be subsets of A . Is it true that $f(X \cup Y) = f(X) \cup f(Y)$? Is it true that $f(X \cap Y) = f(X) \cap f(Y)$? Either prove or give a counterexample in each case.

Proof of 14(a). **Firstly, we want to prove $f(X \cup Y) = f(X) \cup f(Y)$.** We want to prove this by double inclusion.

Firstly, we want to show that $f(X \cup Y) \subset f(X) \cup f(Y)$. Let A and B be arbitrary functions. Suppose there exists $m = f(k)$ such that $k \in X \cup Y$. Now, we may say that if $k \in X \cup Y$ then if $k \in X$ then $m \in f(X)$ or if $k \in Y$ then $m \in f(Y)$. Then, we may write this as

$$m \in f(X) \text{ or } f(Y)$$

$$m \in f(X) \cup f(Y)$$

Thus, we can say that $f(X \cup Y) \subset f(X) \cup f(Y)$.

Now, we want to show that $f(X) \cup f(Y) \subset f(X \cup Y)$. Suppose $m \in f(X)$. Then, there exists a $k \in X$ such that $m = f(k)$. Therefore, by the definition of the union set, we may say that $m = f(k)$ for some $k \in X \cup Y$. Alternatively, suppose $m \in f(Y)$. Then, there exists a $k \in Y$ such that $m = f(k)$. Therefore, by the definition of the union set, we may say that $m = f(k)$ for some $k \in X \cup Y$. As in both cases of $m = f(k)$, $k \in X \cup Y$, we may say that $m \in f(X \cup Y)$. Thus, we may say that $f(X) \cup f(Y) \subset f(X \cup Y)$.

Henceforth, by double inclusion we have proven that the union is true.

Secondly, we will to prove $f(X \cap Y) \neq f(X) \cap f(Y)$. We want to prove this by counterexample.

Let $A = \{1, 2\}$ and $B = \{b\}$. As X and Y are subsets of A , suppose that $X = \{1\}$ and $Y = \{2\}$.

Therefore, by the definition of the sets, we may write $X \cap Y = \emptyset$ and therefore

$$f(X \cap Y) = \emptyset$$

. Now, as previously defined, we may write $f(X) = b$ and $f(Y) = b$. Therefore,

$$f(X) \cap f(Y) = b$$

Thus, this works as a counterexample to the statement that $f(X \cup Y) = f(X) \cap f(Y)$. $\square$

- (b) Now let X and Y be subsets of B . Is it true that $f^{-1}(X \cup Y) = f^{-1}(X) \cup f^{-1}(Y)$? Is it true that $f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$? Either prove or give a counterexample in each case.

Proof of 14(b). **Firstly, we want to prove** $f^{-1}(X \cup Y) = f^{-1}(X) \cup f^{-1}(Y)$. We want to prove this by double inclusion.

Firstly, we want to show that $f^{-1}(X \cup Y) \subset f^{-1}(X) \cup f^{-1}(Y)$. Let A and B be arbitrary functions. Suppose $m \in f^{-1}(X \cup Y)$. Therefore, we may say that $f(m) \in X \cup Y$. Now, we may say that if $f(m) \in X$ then $m \in f^{-1}(X)$. Therefore, we can say that $m \in f^{-1}(X) \cup f^{-1}(Y)$. Alternatively, if $f(m) \in Y$ then we may say that $m \in f^{-1}(Y)$. By these two statement, we may say that $m \in f^{-1}(X) \cup f^{-1}(Y)$.

Thus, we can say that $f^{-1}(X \cup Y) \subset f^{-1}(X) \cup f^{-1}(Y)$.

Now, we want to show that $f^{-1}(X) \cup f^{-1}(Y) \subset f^{-1}(X \cup Y)$. Suppose $m \in f^{-1}(X)$. Then, we may say that $f(m) \in X$ and hence $f(m) \in X \cup Y$. From this, we may then say $m \in f^{-1}(X \cup Y)$. Alternatively, suppose $m \in f^{-1}(Y)$. Then, we may say that $f(m) \in Y$ and hence $f(m) \in X \cup Y$. From this, we may then say $m \in f^{-1}(X \cup Y)$. As in both cases $m \in f^{-1}(X \cup Y)$, we may definitively say that $m \in f^{-1}(X \cup Y)$. Thus, we may say that $f^{-1}(X) \cup f^{-1}(Y) \subset f^{-1}(X \cup Y)$.

Henceforth, by double inclusion we have proven that the union is true.

Secondly, we want to prove that $f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$. We will prove this by double inclusion.

Firstly, we want to prove that $f^{-1}(X \cap Y) \subset f^{-1}(X) \cap f^{-1}(Y)$. Suppose there exists a $m \in f^{-1}(X \cap Y)$. Then we know that $f(m) \in X \cap Y$ and therefore $f(m) \in X$ and $f(m) \in Y$. From this, we may say that $m \in f^{-1}(X)$ and in $f^{-1}(Y)$. Now, we may write this as $m \in f^{-1}(X) \cap f^{-1}(Y)$. Thus, we have that $f^{-1}(X \cap Y) \subset f^{-1}(X) \cap f^{-1}(Y)$.

Secondly, suppose there exists a $m \in f^{-1}(X) \cap f^{-1}(Y)$. From this we may say that $m \in f^{-1}(X)$ and $m \in f^{-1}(Y)$ and therefore $f(m) \in X$ and $f(m) \in Y$. From this, we may say that $f(m) \in X \cap Y$ and thus $m \in f^{-1}(X \cap Y)$. Thus, we have that $f^{-1}(X) \cap f^{-1}(Y) \subset f^{-1}(X \cap Y)$.

Henceforth, by double inclusion we have proven that the statement is true. □

12. Suppose that $A, B \subset \mathbb{N}$. Prove that if A and B are finite and there is a bijection $f : A \rightarrow B$, then $|A| = |B|$.

Proof of AE 13. We want to prove that if A and B are finite then there exists a bijection $f : A \rightarrow B$ then $|A| = |B|$. We want to prove this by direct proof.

Firstly, we want to prove that A and B are finite. As $A, B \subset \mathbb{N}$, by **Theorem 1.38** we may say that A and B are countable. As they are countable, we may assume that they are finite.

Secondly, we want to show that as they are finite, if $f : A \rightarrow B$ is bijective then $|A| = |B|$. We want to prove this by contradiction. As A and B are finite, we may let $A \simeq [a]$ and $B \simeq [b]$ in bijective correspondence. Suppose $|A| \neq |B|$, we may then say that there does not exist an injective function $f : [a] \rightarrow [b]$. □

13. Prove that there is no set A with maximal cardinality. In other words, show that there does not exist a set A with the property that if B is a set, then B has smaller cardinality than A .
14. a) Let X be a countable set and Y a finite set such that $X \cap Y = \emptyset$. Show that $X \cup Y$ is countable.

Proof. Since X is countable, we can define a function f such that $f : X \rightarrow \mathbb{N}$. As X is countable, we know f is a bijection. Furthermore, suppose there exists a set Y that is defined as $Y = \{y_1, y_2, \dots, y_m\}$. Let Y be a finite set.

Now, we may define a function $g : X \cup Y \rightarrow \mathbb{N}$ such that

$$g(x) = \begin{cases} n + i & \text{if } x = y_i \in Y, \\ f(x) & \text{if } x \in X, \end{cases}$$

where $\max(f(X))$ is the largest value of f on X (or 0 if X is finite but empty)

Now, we want to show that the function g is injective.

Case 1: $x_1, x_2 \in X$.

Suppose $x_1, x_2 \in X$ and that $g(x_1) = g(x_2)$. Then, by the definition of g , we have $f(x_1) = f(x_2)$. Since f is injective on X , it follows that $x_1 = x_2$. Therefore, we may say that g is injective on X .

Case 2: $x_1, x_2 \in Y$.

Suppose $x_1, x_2 \in Y$ and that $g(x_1) = g(x_2)$. By the definition of g , we have $g(x_1) = \max\{f(X)\} + i_1$ and $g(x_2) = \max\{f(X)\} + i_2$, where i_1 and i_2 are the indices of x_1 and x_2 in Y . Then $g(x_1) = g(x_2)$ implies $\max(f(X)) + i_1 = \max(f(X)) + i_2$, so $i_1 = i_2$. Since the indices in Y are distinct, this implies $x_1 = x_2$. Therefore, g is injective on Y .

Case 3: $x_1 \in X$ and $x_2 \in Y$.

Suppose $x_1 \in X$ and $x_2 \in Y$. By the definition of g , we have $g(x_1) = f(x_1) \leq \max(f(X))$ and $g(x_2) = \max(f(X)) + i_2 > \max(f(X))$. Therefore, $g(x_1) \neq g(x_2)$. Hence g is injective between elements of X and Y .

Thus, this works to prove that the existence of an injective function from $X \cup Y$ to $\mathbb{N}$. Hence $X \cup Y$ is countable. $\square$

b) Prove that a union of 2 disjoint countable sets is countable.

Proof. Let A and B be disjoint countable sets. Since A is countable, suppose there exists a bijection $f : A \rightarrow \mathbb{N}$. Since B is countable, suppose there exists a bijection $g : B \rightarrow \mathbb{N}$.

Suppose there exists a function h such that $h : A \cup B \rightarrow \mathbb{N}$. We may define h as

$$h(x) = \begin{cases} 2f(x) & \text{if } x \in A, \\ 2g(x) + 1 & \text{if } x \in B. \end{cases}$$

Now, we want to show that this function is injective. Consider the following three cases:

Case 1: $x_1, x_2 \in A$.

Suppose $h(x_1) = h(x_2)$. By the definition of h , this gives

$$2f(x_1) = 2f(x_2) \implies f(x_1) = f(x_2).$$

Since f is injective on A , it follows that $x_1 = x_2$. Thus, h is injective when restricted to A .

Case 2: $x_1, x_2 \in B$.

Suppose $h(x_1) = h(x_2)$. By the definition of h , this gives

$$2g(x_1) + 1 = 2g(x_2) + 1 \implies g(x_1) = g(x_2).$$

Since g is injective on B , it follows that $x_1 = x_2$. Thus, h is injective when restricted to B .

Case 3: $x_1 \in A$ and $x_2 \in B$.

Then $h(x_1) = 2f(x_1)$ is even, while $h(x_2) = 2g(x_2) + 1$ is odd. An even number can never equal an odd number, so $h(x_1) \neq h(x_2)$. Thus, h is injective across A and B .

As we have prove that h is injective in all these cases, we may make the claim that h is always injective. Thus, this works to prove that $A \cup B$ is countable. $\square$

c) Use a) and b) to prove that $\mathbb{Z}$ is countable.

Proof. Let $\mathbb{N}_0 = \{-1, -2, -3, \dots\}$ be the set of negative integers. Then $\mathbb{N}_0$ is countable, since the function $f : \mathbb{N}_0 \rightarrow \mathbb{N}$ defined by $f(n) = -n$ is a bijection.

Now, we may define $\mathbb{Z}$ as $\mathbb{Z} = \mathbb{N} \cup \mathbb{N}_0 \cup \{0\}$.

By part (a), the union of the disjoint sets $\mathbb{N}_0$, a countable set, and $\{0\}$, a finite set, is countable. Then by part (b), the union of the disjoint countable sets $\mathbb{N}$ and $\mathbb{N}_0 \cup \{0\}$ is countable.

Hence, we can conclude that $\mathbb{Z}$ is countable. $\square$

d) Suppose that $A_1, A_2, \dots$ are sets such that A_i is countable for each $i \in \mathbb{N}$. Show that, for each n ,

$$\bigcup_{i=1}^n A_i$$

is countable.

Proof. We prove the statement by induction on n .

Suppose that $A_1, A_2, \dots$ are sets such that A_i is countable for each $i \in \mathbb{N}$

Firstly, let us assess the base case. Let $n = 1$. If $n = 1$, then $\bigcup_{i=1}^1 A_i = A_1$. By assumption, A_1 is countable. Hence the base case holds.

Now, let us assume that for some $k \geq 1$, the union of k countable sets is countable; that is,

$$\bigcup_{i=1}^k A_i \text{ is countable.}$$

Consider $k + 1$ countable sets $A_1, A_2, \dots, A_{k+1}$. We may then write

$$\bigcup_{i=1}^{k+1} A_i = \left(\bigcup_{i=1}^k A_i \right) \cup A_{k+1}.$$

By the inductive hypothesis, $\bigcup_{i=1}^k A_i$ is countable. Since A_{k+1} is countable, and the two sets are disjoint, part (b) implies that the union

$$\left(\bigcup_{i=1}^k A_i \right) \cup A_{k+1}$$

is countable.

Therefore, by the principle of mathematical induction, it follows that for all $n \in \mathbb{N}$, the union

$$\bigcup_{i=1}^n A_i$$

is countable. □

- e) Prove that a countable union of countable sets is countable. That is, if $\{A_i\}_{i \in \mathbb{N}}$ is a countable collection of countable sets, then

$$\bigcup_{i \in \mathbb{N}} A_i$$

is countable.

Proof. Let the countable sets A_i be indexed by $i \in \mathbb{N}$, and enumerate each A_i as $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$. Then, let every element in the union $M = \bigcup_{i \in \mathbb{N}} A_i$ be written as a_{ij} , the j th element of A_i .

Now, let us define f such that $f : M \rightarrow \mathbb{N} \times \mathbb{N}$ by $f(a_{ij}) = (i, j)$. Since $\mathbb{N} \times \mathbb{N}$ is countable, it suffices to show that f is injective.

Case 1: $m, m' \in A_i$ with $m \neq m'$. Then $m = a_{ij}$ and $m' = a_{ij'}$ for some $j \neq j'$. Thus $f(m) = (i, j) \neq (i, j') = f(m')$.

Case 2: $m \in A_i, m' \in A_{i'}$ with $i \neq i'$. Then $m = a_{ij}$ and $m' = a_{i'j'}$. Since $i \neq i'$, we have $f(m) = (i, j) \neq (i', j') = f(m')$.

Therefore, f is injective. Since $\mathbb{N} \times \mathbb{N}$ is countable, M is countable. □

- f) Prove that if A and B are countable then $A \times B$ is countable.

Proof. Since A and B are countable, we can enumerate their elements as $A = \{a_1, a_2, a_3, \dots\}$ and $B = \{b_1, b_2, b_3, \dots\}$ respectively. Then, define a function f such that $f : A \times B \rightarrow \mathbb{N} \times \mathbb{N}$ by $f(a_i, b_j) = (i, j)$.

Now, we want to prove that f is injective.

Suppose $(a_i, b_j), (a_{i'}, b_{j'}) \in A \times B$ and $f(a_i, b_j) = f(a_{i'}, b_{j'})$. By the definition of f , this means that

$$(i, j) = f(a_i, b_j) = f(a_{i'}, b_{j'}) = (i', j').$$

By comparing ordered pairs, we may conclude that $i = i'$ and $j = j'$.

Since the enumerations of A and B are injective, a_i is the i th element of A and $a_{i'}$ is the i' th element of A . Because $i = i'$, we must have $a_i = a_{i'}$. Similarly, $b_j = b_{j'}$.

Therefore, we know that $(a_i, b_j) = (a_{i'}, b_{j'})$. Thus, we can conclude that f is injective.

Since $\mathbb{N} \times \mathbb{N}$ is countable, there exists a bijection $g: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. We have already defined an injective function $f: A \times B \rightarrow \mathbb{N} \times \mathbb{N}$ by $f(a_i, b_j) = (i, j)$.

Now consider the composition $g \circ f: A \times B \rightarrow \mathbb{N}$. Since f is injective and g is bijective, the composition $g \circ f$ is also injective.

Thus, we have an injection from $A \times B$ into $\mathbb{N}$, which by definition means that $A \times B$ is countable. $\square$

g) Prove that if $A_1, A_2, \dots, A_n$ are countable, then so is $A_1 \times A_2 \times \dots \times A_n$.

Proof. We prove the statement by induction on n .

Firstly, let there be a base case of $n = 2$. By part (f), we know that $A_1 \times A_2$ is countable. Hence, this proves the base case.

Now, assume that for some $k \geq 2$, the product $A_1 \times \dots \times A_k$ is countable. We want to prove that $(A_1 \times \dots \times A_k) \times A_{k+1}$ is countable.

Since $A_1 \times \dots \times A_k$ is countable by the inductive hypothesis and A_{k+1} is countable, part (f) implies that $(A_1 \times \dots \times A_k) \times A_{k+1}$ is countable.

Therefore, by induction, it follows that $A_1 \times \dots \times A_n$ is countable for all $n \geq 2$. Thus, this works to prove that $A_1 \times \dots \times A_n$ is countable. $\square$

h) Let $A_n = \{0, 1\}$, for every n . Show that $A_1 \times A_2 \times \dots \times A_n \times \dots$ is uncountable.

Proof. Each element of $A_1 \times A_2 \times \dots$ is an infinite sequence of 0s and 1s. We want to prove that $A_1 \times A_2 \times \dots \times A_n \times \dots$ is uncountable. We will show this by contradiction.

Assume, for contradiction, that all such sequences can be listed as $s_1, s_2, s_3, \dots$, where each $s_n = (s_{n1}, s_{n2}, s_{n3}, \dots)$.

Now, we may construct a new sequence $t = (t_1, t_2, t_3, \dots)$ by defining

$$t_n = \begin{cases} 1 & \text{if } s_{nn} = 0, \\ 0 & \text{if } s_{nn} = 1. \end{cases}$$

We now show that t is not equal to any sequence in the list. Two infinite sequences are equal if and only if all of their coordinates agree. Thus, to prove $t \neq s_n$, it suffices to find one coordinate where they differ.

Fix any $n \in \mathbb{N}$. The n^{th} coordinate of s_n is s_{nn} , while the n^{th} coordinate of t is t_n . By the definition of t_n , we have

$$t_n = \begin{cases} 1 & \text{if } s_{nn} = 0, \\ 0 & \text{if } s_{nn} = 1, \end{cases}$$

so in either case $t_n \neq s_{nn}$. Therefore, t and s_n differ in the n^{th} coordinate, which implies

$$t \neq s_n.$$

Since this argument holds for every $n \in \mathbb{N}$, the sequence t is not equal to any sequence in the list

$$s_1, s_2, s_3, \dots$$

However, t is itself a valid element of $A_1 \times A_2 \times \dots$, so it should have appeared somewhere in the list. This contradiction shows that such a list cannot exist.

Hence, this works to prove that $A_1 \times A_2 \times \dots$ is uncountable. □

18. B is a finite set and A is a subset of B then A is finite. (Be careful about using 1.35 in the proof of (2): 1.35 explicitly assumes both sets are finite!)

Proof. Let B be a finite set. Then, by definition, for some $n \in \mathbb{N}$ there exists a bijection

$$f : B \rightarrow \{1, 2, \dots, n\}$$

Let $A \subseteq B$. To show that A is finite, it suffices to construct a bijection between A and some subset of $\{1, 2, \dots, n\}$. Therefore, define the set

$$S = \{i \in \{1, 2, \dots, n\} : f(i) \in A\}.$$

Since S is a subset of the finite set $\{1, 2, \dots, n\}$, the set S is finite. We now show that f gives a bijection between S and A .

First, if $i \in S$, then $f(i) \in A$ by definition of S , so $f(S) \subseteq A$. Conversely, if $a \in A$, then $a \in B$, so since f is bijective, there exists a unique $i \in \{1, 2, \dots, n\}$ with $f(i) = a$; and because $a \in A$, that index i lies in S . Thus $A \subseteq f(S)$.

Hence $f(S) = A$. Since f is injective on all of $\{1, 2, \dots, n\}$, it is injective on S , and therefore the map

$$f : S \rightarrow A$$

is a bijection.

Thus A is in bijection with the finite set S , and therefore A is finite. □